

Looking Ahead in Goal-Oriented Dialogue

Preference-Tree (PTO) vs Group-Relative (GRPO) optimization of a small Motivational-Interviewing therapist

Project status — all four arms complete, both graders, the compute axis, and a $K \times$ method answer

Every arm is trained and fully scored. Look-ahead's value turns out to be method-dependent — and visible mostly to the grader that was NOT the reward.

Lior Baruch · Reichman University · 18 August 2026

Llama-3.2-1B therapist (bf16) · gpt-4o-mini simulated patient + oracle · Claude Haiku 4.5 held-out judge · 96 personas, persona-paired

Where the project stands, and what I'd like to decide

Settled

- **All four arms are finished and fully scored.** PTO and GRPO at $K \in \{0,5\}$, 39 model states, every conversation scored on eight instruments by two independent graders — 29,952 cells per grader ($39 \times 8 \times 96$).
- **The comparison now has a COST axis.** GPU-hours per iteration, reconstructed from run artifacts. PTO reaches iteration 10 for 3.4× less compute than GRPO — and the two GRPO arms turn out to be budget-matched within 3%.
- **The look-ahead question is answered, per method.** On GRPO, $K=5$ wins — reward, MI-consistency at matched budget, and how much of the gain a held-out judge credits. On PTO it closes the flattery channel and buys nothing else.
- **Both drafts were revised today** after a table-first audit; both compile at or under the 8-page body limit with complete related-work sections.

Open — the agenda

- **Replication.** Every contested endpoint is a single 96-conversation draw. A second draw from 5 adapters costs ~\$11 and either retires the objection or changes the headline. Go?
- **A third grader.** The one purchase that would break the substitution paper's central tie (the aggregate the two judges disagree on).
- **Scope and order.** Which draft is finished first; what the thesis chapter is built around.
- **Human MI coders · authorship · venue** — unchanged from last time, still open.

In one line: the therapist learns to flatter; trajectory-level scoring stops the flattery and the therapist advises instead; which method you optimise with decides whether look-ahead pays — and only the grader outside the training loop can see most of it.

The starting point — the ICLR 2025 paper

Preference Tree Optimization: Enhancing Goal-Oriented Dialogue with Look-Ahead Simulations

Baruch, Butman, Bar, Friedman · ICLR 2025 (SSI-FM workshop poster)

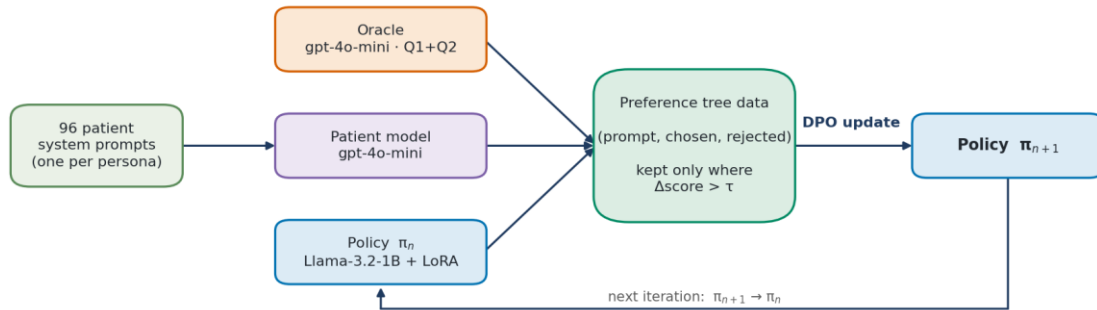
- **The method.** At each therapist turn, branch several candidate replies; for each, simulate **K further turns**; let the oracle score the resulting trajectory; keep best and worst as a preference pair; update with DPO. Repeat, regenerating from the improved model.
- **Why look ahead.** Scoring a reply on its own rewards openings that look good in isolation. Scoring the reply plus K simulated turns rewards openings that lead somewhere.
- **Headline finding.** Every PTO model beat the untrained baseline, and **K = 5 scored higher and more stably than K = 0** — on a 7B therapist over 7 iterations.
- **Why that matters today.** On a 1B model over 10 iterations, that result reproduces on one optimiser and not the other — and where it reproduces, most of the effect is visible only to a held-out grader. Slides 12–16.

What was run, what it cost, and what is scored

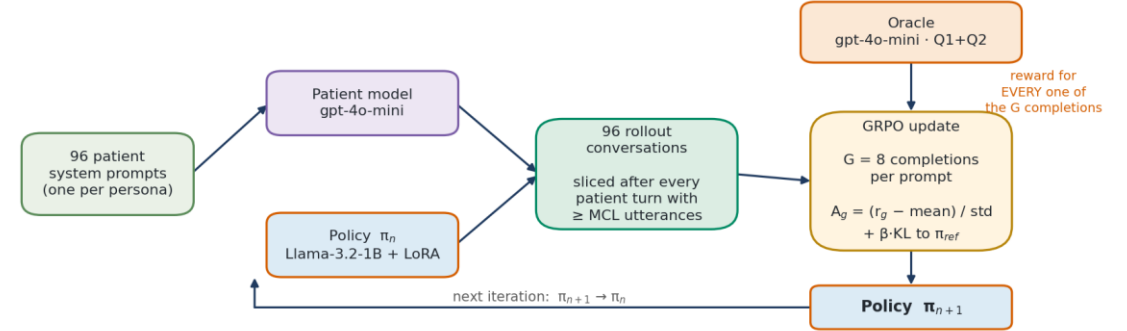
Arm	Trained	Scored (both graders)	GPU-h	Status
PTO K=0	iterations 1–10	base + 1–10	8.1	complete
GRPO K=0	iterations 1–10	base + 1–10	27.9	complete
PTO K=5	iterations 1–10	base + 1–10	19.7	complete
GRPO K=5	iterations 1–5	base + 1–5	27.1	complete — budget-matched to K=0

- **39 scored model states** — 29,952 cells per grader ($39 \times 8 \times 96$), complete on both, no partial cells.
- **“GRPO K=5 stopped at iteration 5” is a statement about iteration count, not spend** — 27.08 vs 27.91 GPU-h is a 3% difference. A K=5 optimizer step costs $\sim 1.9\times$ a K=0 step (measured per iteration), so matched-iteration tables hand K=5 double the compute per cell. Every K contrast in this deck is read on BOTH axes.
- **Matched by construction** across every arm: sampling temperature 1.2, 8 candidates per branch point, minimum context length 12, the same τ filter, the same DPO β , the same oracle. Verified against each run's stored metadata, not assumed.
- **Training reward = two rubrics** (session satisfaction + working alliance). The other six instruments were never optimised against, which is what makes them evidence. All comparisons are persona-paired on the recovered persona identity.

The two methods — same loop, one step different



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
 The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.
 As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

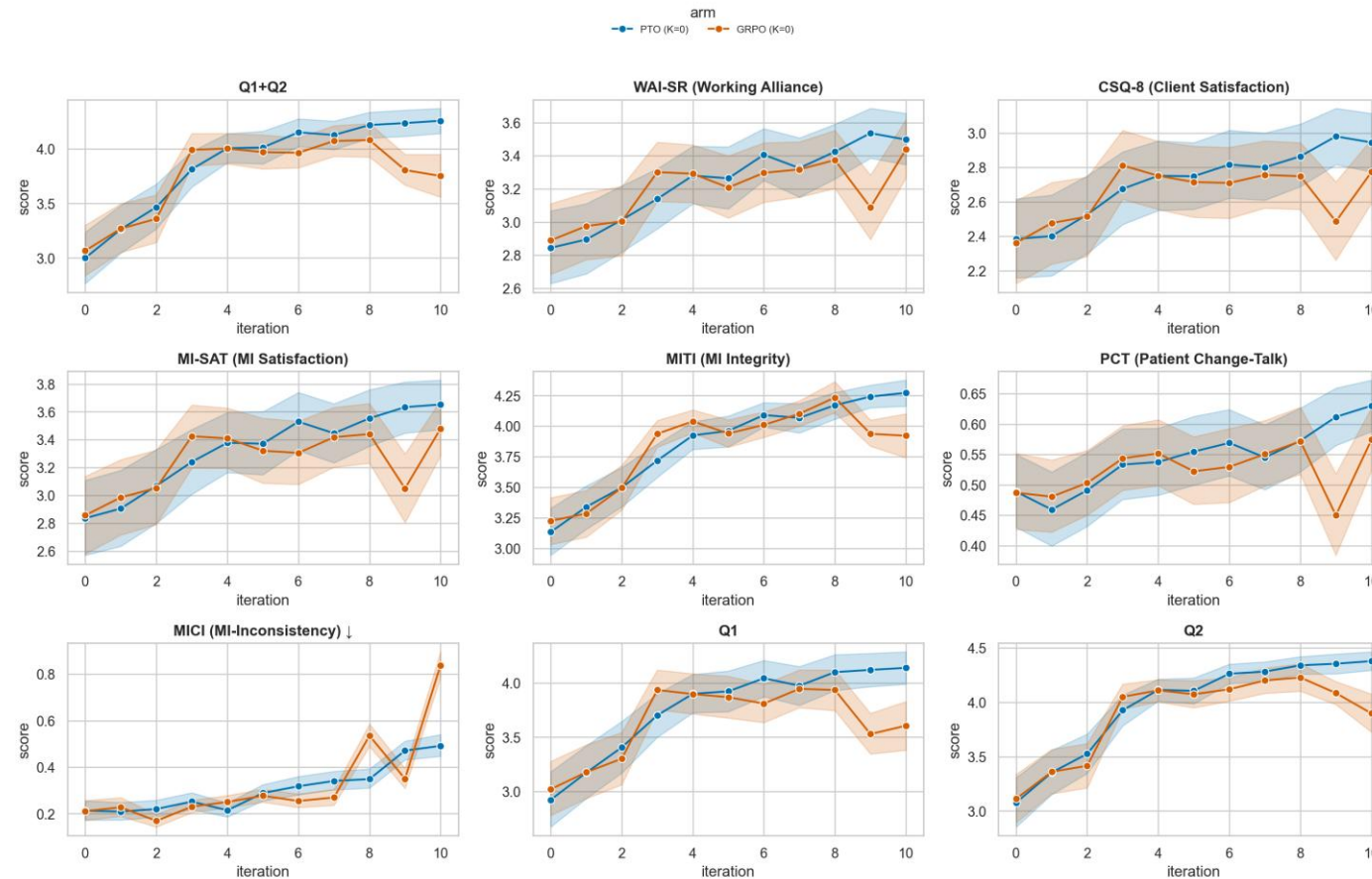
PTO — branch, score, keep best vs worst, DPO

GRPO — sample a group, standardize rewards, policy gradient

Both regenerate their own training data from the current policy each iteration. Look-ahead (K) changes only what CONTEXT the oracle grades — never the loss, the hyperparameters, or the sampling.

All eight instruments across 10 iterations (K = 0)

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor, PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

Mean $\pm$ 95% CI over the 96 personas, primary oracle. MICI is lower-is-better. PTO climbs steadily to iteration 10; GRPO peaks at iteration 8 and regresses.

Endpoint numbers (K = 0)

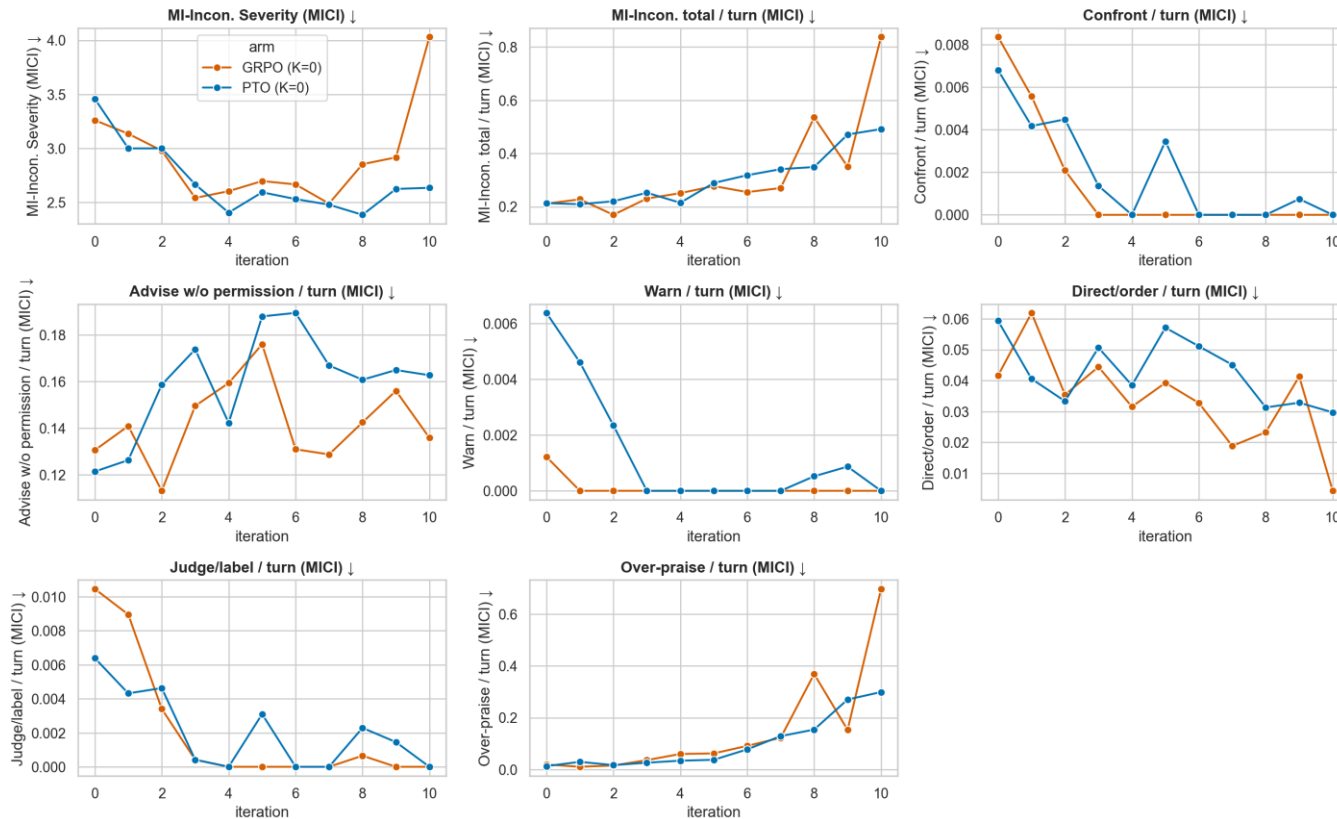
Primary oracle, final and best endpoint. Read off the generated scorecard (results/L0/tables/0_headline/gpt-4o-mini/) — every cell traces to the score lake.

	Q1+Q2	MITI	MICI ↓	R:Q	% MI-consistent
PTO K=0 @ iter 10	4.26	4.27	0.49	0.75	0.70
GRPO K=0 @ iter 10	3.75	3.92	0.84	1.44	0.83
GRPO K=0 @ iter 8 (its peak)	4.08	4.23	0.54	1.04	0.71

- **PTO beats GRPO at the matched endpoint** — Q1+Q2 4.26 vs 3.75, persona-paired +0.51, dz 0.73.
- **Credit GRPO at its peak instead** and PTO still leads (+0.18, dz 0.30, p = .010) — but the MITI and MICI gaps stop being significant. **We claim the weaker version.**
- **GRPO's higher R:Q is not a win.** Its reflection-to-question ratio crosses the MITI competency bar because it stopped asking questions — the regex question rate falls 0.83 → 0.15 per turn. A ratio is gameable through its denominator.
- **“Best” is itself grader-dependent.** The held-out judge orders the arms the same way (PTO@10 2.87 vs GRPO@10 2.26) — but it puts GRPO's best at iteration **3**, not 8. Peak selection performed on the very signal being optimised is not a neutral operation. Never average the two graders' raw scores — this is train-vs-test, and only contrasts combine.

What the model actually learned (K = 0)

MICI detail — severity + per-therapist-turn rates (higher = worse)



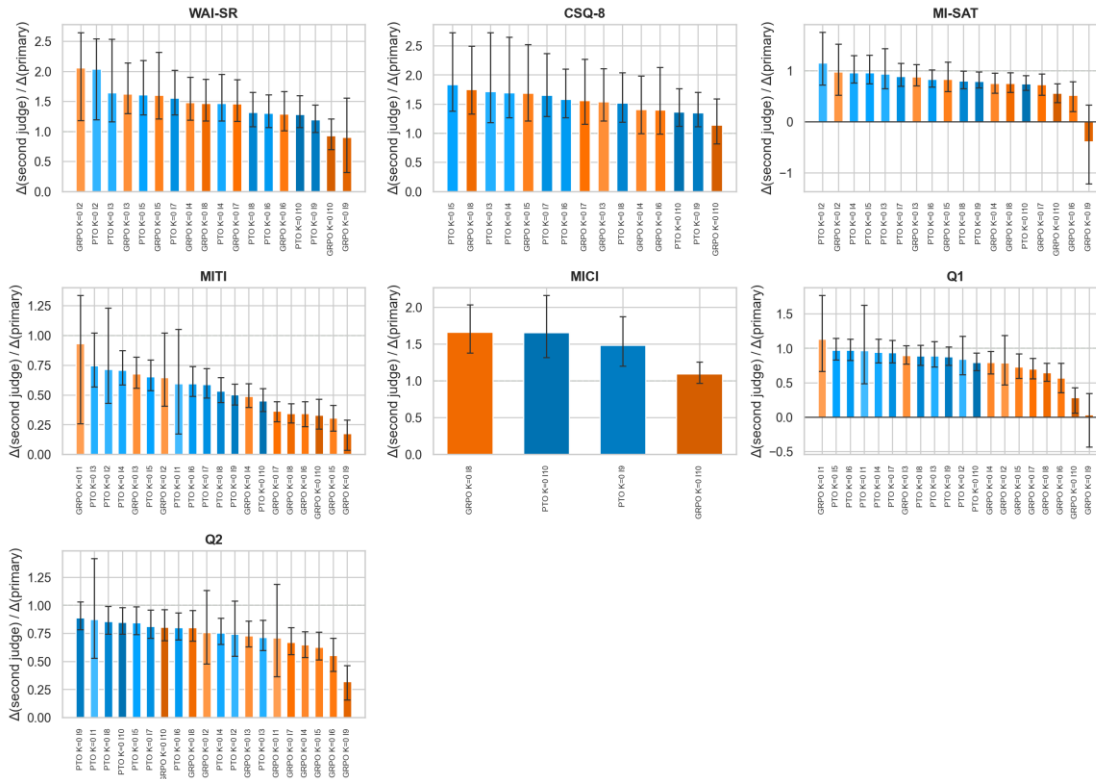
MI-inconsistent behaviour, decomposed, per therapist turn. Higher is worse.

One channel carries the whole thing

- Every **coercive** MI violation FALLS with training — confronting, warning, directing and judging all go to ~0.
- Over-praise rises from **0.02 to 0.70 per turn** in GRPO and 0.01 to 0.30 in PTO, and accounts for more than 100% of the increase in total MI-inconsistency.
- The rubrics measure how the session **felt to the client**. Flattery is the cheapest way to move that.
- So the model did not get broadly worse at MI. It swapped coercion for flattery.

How much of this survives a grader that took no part in training

[EVAL] Does the improvement transfer? Gain over PTO K=0 Base as seen by Claude Haiku 4.5, relative to the primary oracle it was trained against (dashed = full transfer)



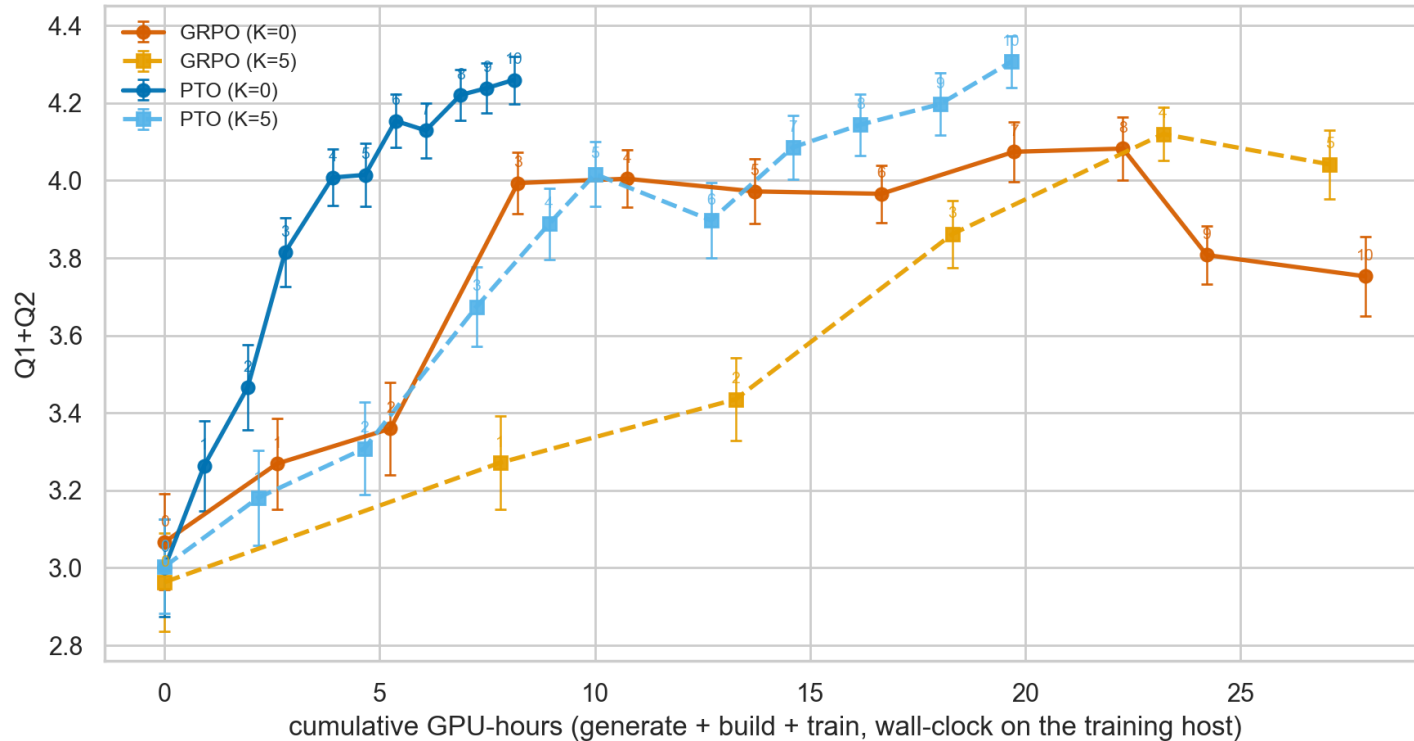
Fraction of each arm's gain over base that survives the judge swap. ~1.0 = a real behaviour change; ~0 = a gain that existed only in the optimised grader.

Gain retention = a train/test split on the reward

- The primary oracle **was the training signal**; Claude Haiku 4.5 is held out, from another family, and never played the patient. Agreement between them is a generalisation check, not inter-rater reliability.
- Oracle repeatability ICC(2,1) **0.86–0.99**; sign preservation across all arm × metric contrasts rises to 98.9% at $|\Delta| \geq 0.50$.
- Q1 retention: PTO@10 **0.80** vs GRPO@10 **0.28**, non-overlapping — GRPO's late gains are increasingly grader-specific. The retention curve separates the arms from iteration 4, four iterations before the reward curve turns over.
- △ Standing caveats: MITI dependability is **0.65** off one judge, and MICI agreement is weak at the conversation level. Both are limitations in the write-up, not footnotes.

The compute axis — an iteration is not a unit of spend

Q1+Q2 against COMPUTE, not iteration
marker labels are iteration numbers



The primary metric against cumulative GPU-hours instead of iteration; iteration numbers annotated on the markers.
Unequal marker spacing IS the finding.

Reconstructed from run artifacts, not from logs

- PTO reaches iteration 10 for **8.1 GPU-h** vs **GRPO's 27.9** — 3.4× cheaper, and it scores higher. Its preference-tree build runs once per iteration; GRPO recomputes its reward inside the training loop on every step.
- But at MATCHED spend (~8 GPU-h, GRPO only at iteration 3), PTO@10 wins the reward (+0.27, dz 0.53) while being **worse on MI-inconsistency** (+0.26, dz 0.90; both replicate held-out): the hack tracks optimization depth, so more reward per GPU-hour buys more reward-hacking per GPU-hour.
- For look-ahead, the lever's sign is a **function of budget**: K=5 (GRPO) is clearly worse at ~13 GPU-h (dz -0.74), null at 7.8 and 23–27 on the primary oracle — and ahead at the top budgets only under the held-out judge (dz +0.31–0.33).
- **Quote the curve, never one row** — and name the axis (iteration vs budget) on every K claim; the MICI contrast reverses sign between them.

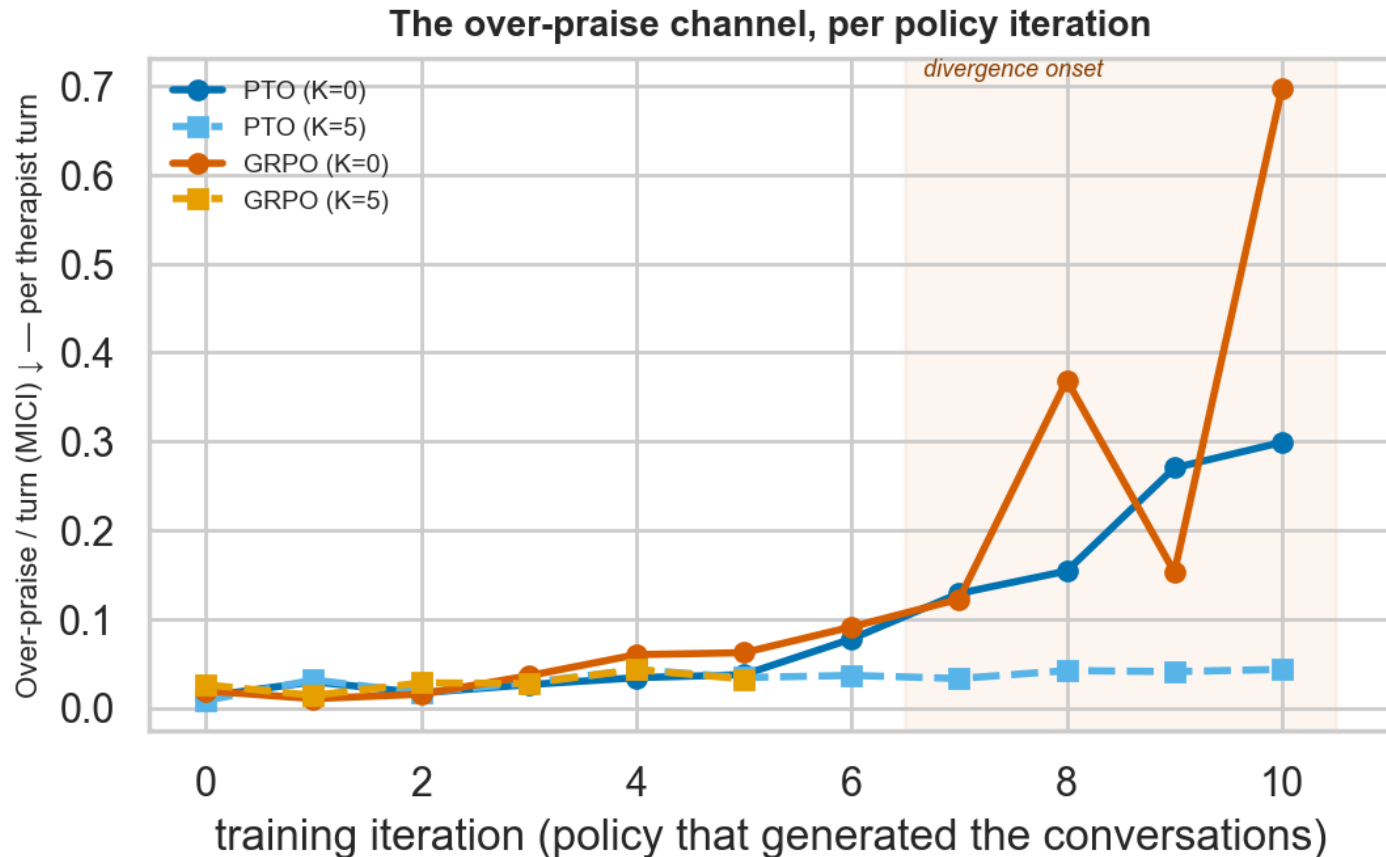
Now a $K \times$ method comparison

Does look-ahead fix the reward hack?

It closes the channel it targets, on both methods. What else it buys depends on the optimiser — and on who is grading.

PTO: eleven matched iterations (0–10). GRPO: six (0–5), budget-matched to its $K=0$ arm within 3%. Both graders, persona-paired.

Look-ahead closes the channel it targets, and keeps it closed



Over-praise per therapist turn, all four arms. K=0 solid, K=5 dashed. Shading marks the measured divergence onset.
Higher is worse.

Exactly what it was supposed to do

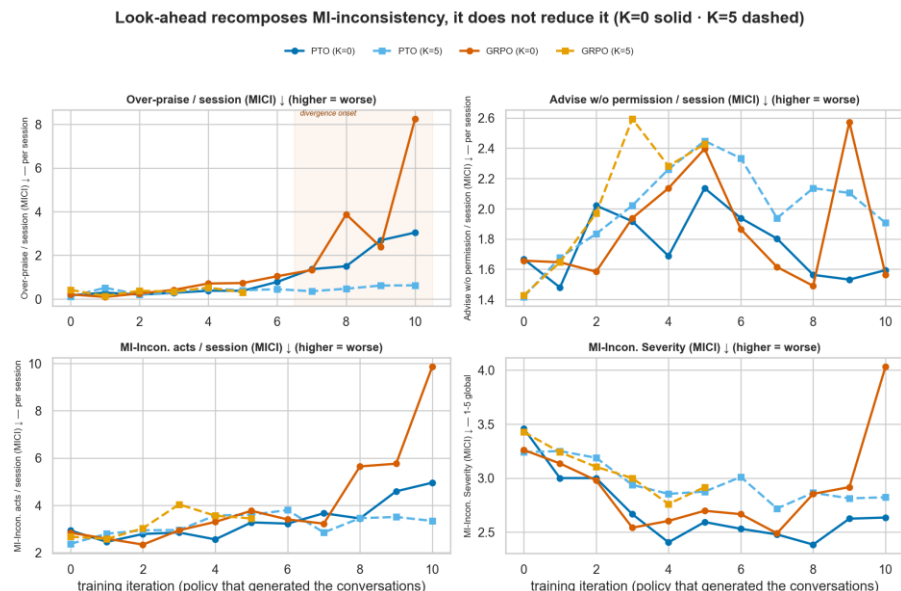
- The two PTO arms are **indistinguishable for six iterations** and then separate for good. Over-praise per session at iteration 10: **3.042 (K=0) vs 0.625 (K=5)**.
- Persona-paired, $n = 96$: **dz 0.887, $p < 10^{-4}$** . The held-out judge reports a larger gap in the same direction (dz 0.999).
- Prevention, not delay.** K=5's over-praise stays LOW to the end of the run — 0.115 → 0.625 across eleven points, while K=0 goes 0.167 → 3.042.
- Not a denominator artifact — these are **raw per-session counts**. K=5 takes MORE therapist turns (14.4 vs 10.2), which would push a rate the other way. Not data starvation — K=5 trained on more preference groups (6,416 vs 4,935).
- Replicates on GRPO at its matched iterations: over-praise as a share of MI-inconsistent acts falls 0.178 → 0.086 (primary) and 0.182 → 0.063 (held-out) at iteration 5.

...and the substitution replicates — the reduction does not

per session, iteration 10 (PTO)	Δ (K=0 – K=5)	dz	p (Holm)
Over-praise	+2.417	+0.887	$< 10^{-4}$
Advice without permission	-0.313	-0.239	.057 n.s.
Directing the client	-0.375	-0.389	.0016
ALL MI-inconsistent acts	+1.615	+0.446	.0003

The therapist changed how it fails; whether it fails less depends on who grades

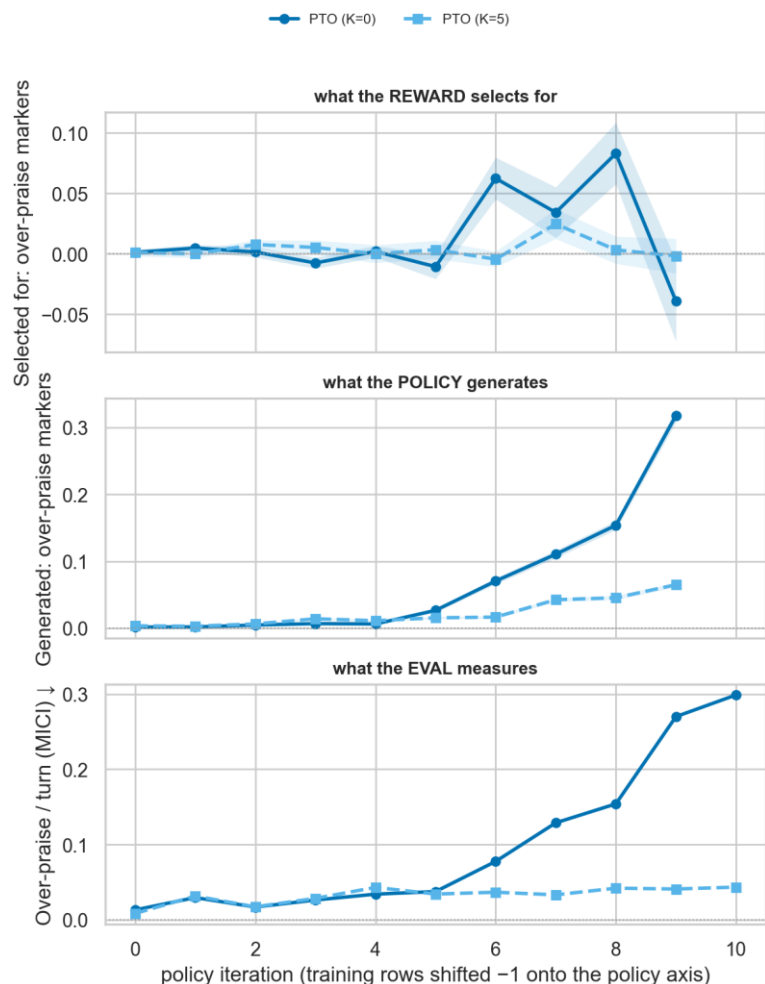
- Table above is the **primary oracle**, which WAS the training reward: totals 4.958 (K=0) vs 3.344 (K=5) acts per session.
- The held-out judge reproduces every component and **larger** — over-praise dz 0.999, advice -0.709, directing -0.468 — but its total is **null: +0.531, dz 0.099, p = .167**.
- So the claim that travels across graders is **substitution, not reduction**. The significant aggregate is a primary-oracle-only result and is labelled as one wherever it appears.
- A second, independently prompted coder agrees on the trade:** in the MITI behaviour counts, **persuade** rises under K=5 by 0.67 acts/session (dz 0.52 primary) and 2.63 (dz 0.87 held-out), while affirmations fall on both measures.
- Counts, not rates.** The per-TURN rate is large under both judges only because K=5 takes 14.4 therapist turns to K=0's 10.2. Per-session counts are the primary reading.
- And the reward the run optimised still cannot separate the arms at the endpoint (dz -0.096, p_holm .695 primary; dz 0.308, p_holm .130 held-out — both null).



Per-session counts — no denominator involved.

Where the intervention acts — and why the pressure relocates

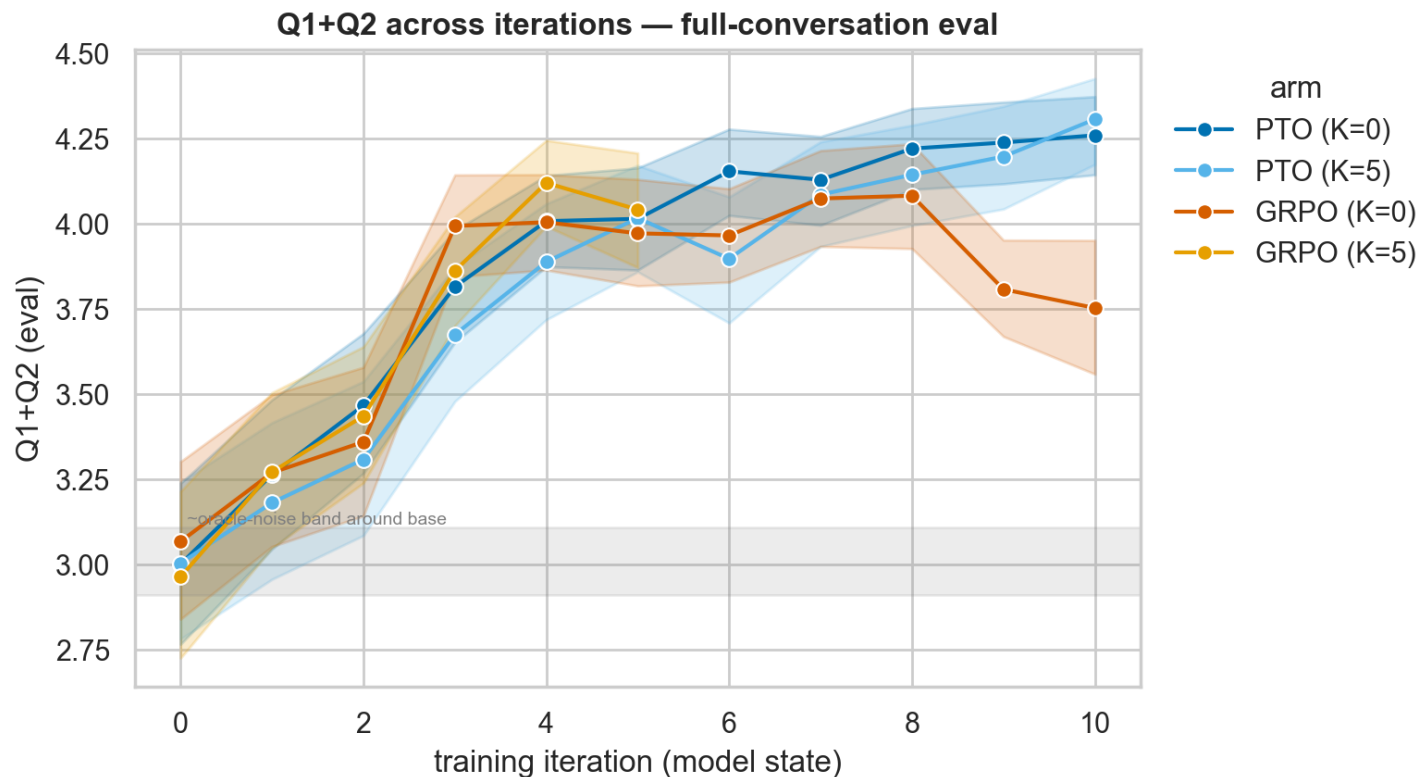
Over-praise: where look-ahead intervenes (PTO, K=0 solid · K=5 dashed)



A behaviour has to clear three gates

- Every training group logs all its candidates and their oracle scores, so we can measure what the reward **selected for** — not just what the policy produced.
- Under K=0 the selection weight on over-praise is ~0 for **six** iterations, then jumps — 0.063, 0.034, **0.083** at policy iterations 6–8 — and the candidate pool and the evaluation follow with a lag.
- Under K=5 it **never clears +0.025** at any iteration; the pool moves 0.004 → 0.065 and the evaluated rate 0.008 → 0.043. Look-ahead acted on the **reward**, not on the policy.
- Which is also why the substitution is unsurprising. The objective did not change; only the price of one route went up. A trajectory-extended oracle still rewards a turn that makes the next five look productive — and **giving advice does that too**. Nothing penalises advice; the intervention penalises praise that does not compound.
- Scale worth holding together: the largest per-iteration selection pressure anywhere in this chain is **0.083**, while the generated over-praise rate moves **0.002 → 0.318**. A bias too small to see in one update compounds through the on-policy loop.

The K × method answer — look-ahead pays on GRPO, not on PTO



Both K arms of both methods, Q1+Q2, mean $\pm$ 95% CI. K=5 lines stop at their last scored iteration (PTO 10, GRPO 5); GRPO's two arms cost the same total GPU-hours.

Same lever, opposite verdicts

- **On PTO, K=5 never significantly leads on the reward** — over eleven matched iterations, under either grader.
- **On GRPO, K=5 leads** — at iteration 4 on both graders (Δ 0.115 primary, 0.233 held-out, both Holm-sig.) and at iteration 5 under the held-out judge (Δ 0.311, dz 0.43).
- **At matched BUDGET the GRPO gain is larger and MICI reverses:** K=5@5 vs K=0@10 (27.1 vs 27.9 GPU-h): Q1+Q2 +0.289 (dz 0.36, primary) / +0.540 (dz 0.84, held-out); MI-inconsistency per turn **-0.497 (dz -1.34) / -0.403 (dz -1.23)** — far cleaner MI at equal spend.
- **Look-ahead FLIPS which method wins at iteration 5** — K=0: PTO leads (+0.265, p .014); K=5: GRPO leads (-0.219, p .005); difference-in-differences dz 0.525, p .0001. **On the primary oracle the same interaction is null** — the grader that WAS the reward cannot see it.

New this week — look-ahead decides how REAL the gains are, per method

Q1 gain retention (held-out ÷ primary)	K = 5	K = 0	read
GRPO @ iteration 5	1.08 [0.94, 1.27]	0.73 [0.57, 0.92]	disjoint — K=5 fully credited
PTO @ iteration 10 (Q1)	0.72 [0.61, 0.84]	0.80 [0.68, 0.93]	overlapping
PTO @ iteration 10 (Q2)	0.56 [0.47, 0.66]	0.85 [0.74, 0.98]	disjoint — K=5 WORSE

- **Under K=5, GRPO's gains are fully credited by the judge that never trained it** — where its K=0 gains were already leaking by iteration 5 and collapsed to 0.28 by 10. The retention-space counterpart of the interaction on the previous slide.
- **For PTO, closing the flattery channel bought no transferability** — the same on Q1, significantly less on Q2. Consistent with substitution: the pressure found another outlet with the same grader-specific character.
- **How this number almost didn't exist:** the K=5 retention table had rendered as a silent 0-byte file (a hardcoded reference model the K=5 view excludes). A table-first audit caught it on 2026-08-18; the export layer now refuses to write a silently-empty artifact. Next slide.

Caveat: cross-view comparison against different draws of the identical base policy; the measured base noise floor (54 same-policy contrasts, max |dz| 0.15) bounds the draw effect. The Q2 gap is far beyond it.

What a table-first audit caught this week — and why it stays on a slide

A silent empty artifact (prose never noticed; the file 'existed')

The L5 gain-retention table was 0 bytes for weeks — the producing notebook hardcoded a reference model the view excludes, and an empty frame serialized to nothing. Fixed, guarded (empty tables now write an explicit marker), and the un-broken table produced slide 16.

A factual claim in the CLPsych draft (a summary hardened into a finding)

§5 said the same three reward items lead the endpoint gains in both arms. The table says self-disclosure tops only GRPO; warmth tops PTO. Corrected, and added to the paper's ledger of claims that are easy to get subtly wrong.

A placeholder citation and a miscited paper (caught before any reviewer could)

One bib entry was literally PLACEHOLDER; another (a conversation-planning paper) was cited for therapist-behaviour analysis. Both verified against sources and re-homed.

- **Same shape as the two corrections presented on 2026-08-16:** prose about tables drifts; the tables themselves have never been wrong. The discipline — read the tables cold, then diff the narrative — is now written into the repo's own rules, and it caught the auditor too (a sign-convention misread, fixed before anything shipped).

Two drafts, deliberately disjoint — both revised today

Affirmation Without Inquiry — $K = 0$ only (submission-ready pending decisions)

What the reward hack IS. Revised 2026-08-18: the §5 item-level claim corrected, the compute axis added to Discussion (3.4× cheaper, hackier at matched budget), the retention figure now a legible single panel, citations verified. Body ends exactly at p.8; 0 overfull boxes, 0 undefined references.

The Hack Moves — PTO only, $K \in \{0,5\}$ (complete draft)

Whether a principled fix removes the hack. Revised 2026-08-18: related work written (was a scaffold), the full channel table in Appendix A — surfacing the MITI-persuade corroboration — the retention result added to §4, reproducibility filled from run metadata. Body at 68% of p.7; ~1.3 columns of slack.

- **They share no claims.** The split is the K axis: the first varies the optimiser at $K=0$; the second varies K within PTO. Each carries a claim→artifact ledger; every number traces to a generated table, which is what caught this week's error.
- **Neither is a $K \times$ method paper — by scope, not by data.** GRPO $K=5$ is now trained 1–5 and scored on both graders, so the slide-15 contrast exists and both drafts say in Limitations that they state their claims outside that window. Where the $K \times$ method result goes — thesis chapter or third paper — is part of Decision 1.

Decision 1 — scope and order

Two drafts exist and the thesis needs a spine. These are not mutually exclusive; the question is what gets finished first and where the $K \times$ method result lives.

Submit the $K=0$ paper first; look-ahead + $K \times$ method become the thesis' core (my read: lowest risk, best fit)

The reward-hacking result is the more complete and self-contained. The look-ahead story — substitution within PTO, the method-dependent payoff, the compute axis — is bigger than 8 pages and is exactly a thesis chapter.

Submit the substitution paper first (higher ceiling, higher risk)

The more novel claim, with a methodological contribution beyond MI (nominate the aggregate before the intervention; composition breaks grader-invariance). One optimiser, one run per arm; its own TODO names a third grader as the tie-breaker.

Fold the $K \times$ method result into a third short paper

GRPO $K=5$'s full-credit retention + the interaction only the held-out grader sees is a clean standalone story — but it competes with the thesis chapter for the same material.

Whatever the order: both drafts are frozen against their ledgers, so finishing either is now a decisions problem (authors, venue, human coders), not an analysis problem.

Decision 2 — what, if anything, is worth buying

Total API spend to date ≈ \$317 and it is the binding constraint. GRPO at K=5 — last meeting's big question — has been bought and settled. What's left:

A second independent draw from 5 contested adapters (~\$11.4 + ~1 A100-h) (recommended)

GRPO_LA0 @3/@8/@10 and GRPO_LA5 @4/@5. Every contested endpoint is a single 96-conversation draw; the only measured noise floor is at the base (54 same-policy contrasts, all null). Either it retires the endpoint-fragility objection thesis-wide, or GRPO's iteration-9/10 collapse fails to reproduce and the headline changes. Decisive both ways.

A third grader on the substitution paper's aggregate (the runner-up)

The two judges agree on every component and disagree on the total. A third family breaks the tie either way and converts the paper's central hedge into a finding. Priced off the receipt-calibrated basis before any go.

Extending GRPO K=5 to iteration 10 (~\$118 + 23–34 A100-h) (declined, with the reason on the slide)

NOT recommended: it would push that arm to ~50 GPU-h against its K=0 twin's 28, making the arms LESS comparable on the axis that currently makes them a clean pair.

What I'd like to leave with

- **A go / no-go on the replicate draw.** ~\$11.4; the single highest-value purchase left, and both papers' framing depends on the endpoint it tests.
- **A decision on order.** Which draft is finished first, and does the $K \times$ method result live in the thesis or in a third paper?
- **A view on the third grader** for the substitution paper's aggregate tie.
- **A view on human MI coders.** Two graders from different families agree, but agreement is not validity, and no MI-trained human has read a single transcript. Even 40–50 coded sessions would change what we can claim — do we have access to a coder?
- **Co-author list and order** for both drafts, plus venue and target date.

Everything in this deck regenerates from one command. Figures, tables and summaries rebuild under either grader with seeded intervals; the analysis package carries a 22-check self-test that runs after every change (22/22 passing as of this deck).